

1)

$$\eta_1 = \sqrt{\frac{\mu_1}{\epsilon_1}} \frac{\eta_0}{\sqrt{\epsilon_{r1}}} = 40\sqrt{3}\pi \Omega$$

$$\Gamma = \frac{\eta_1 - \eta_0}{\eta_1 + \eta_0} = -0.27, \quad (2 \text{分}) \quad T = \frac{2\eta_1}{\eta_1 + \eta_0} = 0.73 \quad (2 \text{分})$$

$$\vec{E}^r = \Gamma \vec{E}^i = -0.27 E_0 (\vec{x}_0 - j\vec{y}_0) e^{jk_z z}, \quad |\vec{E}^r| = 0.27\sqrt{2} E_0 = 0.38 E_0 \quad (2 \text{分})$$

2)

$$\vec{E}^r = 0.73 E_0 (\vec{x}_0 - j\vec{y}_0) e^{-jk_z z}, \quad (2 \text{分})$$

$$\vec{H}^r = \vec{z}_0 \times (\vec{x}_0 - j\vec{y}_0) \frac{0.73 E_0 e^{-jk_z z}}{\eta_1} = 3.4 \times 10^{-3} E_0 (\vec{y}_0 + j\vec{x}_0) e^{-jk_z z} \quad (2 \text{分})$$

$$|\vec{H}^r| = 3.4\sqrt{2} \times 10^{-3} E_0 = 4.8 \times 10^{-3} E_0 \quad \text{其中 } k_2 = \sqrt{3}k \quad (2 \text{分})$$

3) 入射波为右旋圆极化 (2分), 反射波为左旋圆极化 (2分), 透射波为右旋圆极化 (2分)

2. 一个空气填充的矩形波导, 横截面尺寸为 $a \times b$, 其中 $a = 80 \text{ mm}$, $b = 40 \text{ mm}$ 。1) 试求主模 TE_{10} 模下的截止波长; (4分)2) 若工作频率 f 为主模截止频率的 2.5 倍, 试问有哪些模式可以在该波导中传播? (8分)3) 计算在 2) 中工作频率 f 处 TE_{10} 模的波导波长和波阻抗 ($Z_{\text{TE}} = \eta \frac{\lambda_g}{\lambda}$)。 (8分)

1)

$$\lambda_c = \frac{2\pi}{\sqrt{\left(\frac{m\pi}{a}\right)^2 + \left(\frac{n\pi}{b}\right)^2}} = \frac{2}{\sqrt{\left(\frac{m}{a}\right)^2 + \left(\frac{n}{b}\right)^2}} \quad (2 \text{分})$$

$$\text{TE}_{10} \text{ 模的截止波长为 } \lambda_c = 2a = 160 \text{ mm} \quad (2 \text{分})$$

2)

$$f_c = \frac{c}{\lambda_c}, \quad \text{令 } \left(\frac{2\pi \times 2.5 f_c}{c}\right)^2 \geq \left(\frac{m\pi}{a}\right)^2 + \left(\frac{n\pi}{b}\right)^2, \quad (3 \text{分})$$

$$\text{即 } \left(\frac{2\pi \times 2.5}{\lambda_c}\right)^2 \geq \left(\frac{m\pi}{a}\right)^2 + \left(\frac{n\pi}{b}\right)^2$$

满足此关系的波有 TE_{10} , TE_{01} , TE_{11} , TE_{20} , TM_{11} (5分)

3)

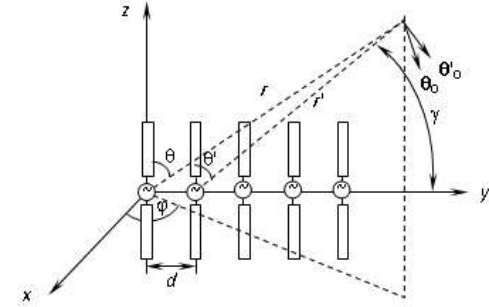
$$\lambda_g = \frac{\lambda}{\sqrt{1 - \left(\frac{\lambda}{\lambda_c}\right)^2}} = \frac{\lambda}{\sqrt{1 - \left(\frac{f_c}{f}\right)^2}} = \frac{\lambda}{\sqrt{1 - \left(\frac{1}{2.5}\right)^2}} = 1.091\lambda \quad (4 \text{分})$$

$$\text{而 } \lambda = c/f = 64 \text{ mm} \quad \text{故 } \lambda_g \approx 70 \text{ mm}$$

$$Z_{\text{TE}_{10}} = \frac{\eta}{\sqrt{1 - \left(\frac{f_c}{f}\right)^2}} = 1.091\eta = 411.3 \Omega \quad (4 \text{分})$$

五、五单元边射阵, 天线元间距为 $\frac{\lambda}{2}$, 各天线元上的电流按 1:2:3:2:1 分布, 试确定阵因子

和归一化方向图。(15分)



解: 如图所示, 设天线 1 在考察点 P 的辐射场为 $\mathbf{E}_1$ 。由于天线 2 上的电流 $I_2 = 2I_1 e^{j\phi}$ (式中 $\phi = kd \cos \phi$), 则天线 2 在 P 点的辐射场为 $\mathbf{E}_2 = 2\mathbf{E}_1 e^{j\phi}$ 。同理, 天线 3、4、5 在 P 点的辐射场分别为 $3\mathbf{E}_1 e^{j2\phi}$, $2\mathbf{E}_1 e^{j3\phi}$, $\mathbf{E}_1 e^{j4\phi}$, 于是 P 点的总辐射场为

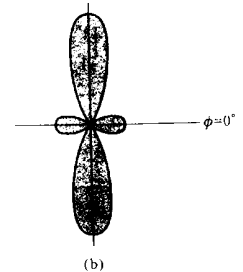
$$\mathbf{E} = \mathbf{E}_1 (1 + 2e^{j\phi} + 3e^{j2\phi} + 2e^{j3\phi} + e^{j4\phi})$$

则五元边射阵的归一化阵因子为

$$|F(\phi)| = \frac{1}{9} |1 + 2e^{j\phi} + 3e^{j2\phi} + 2e^{j3\phi} + e^{j4\phi}| = \frac{1}{9} \left| \frac{1 - e^{j5\phi}}{1 - e^{j\phi}} \right| = \frac{1}{9} \left| \frac{\sin\left(\frac{5\phi}{2}\right)}{\sin\left(\frac{\phi}{2}\right)} \right|^2$$

将 $\phi = kd \cos \phi$ 代入上式, 得

$$|F(\phi)| = \left| \frac{\sin\left(\frac{3}{2}\pi \cos \phi\right)}{3\sin\left(\frac{1}{2}\pi \cos \phi\right)} \right|^2$$



由此可画出五元边射阵的方向图如右图所示。